



United International University (UIU)

IPE 401/IPE3401: Industrial Management/Industrial & Operational Management

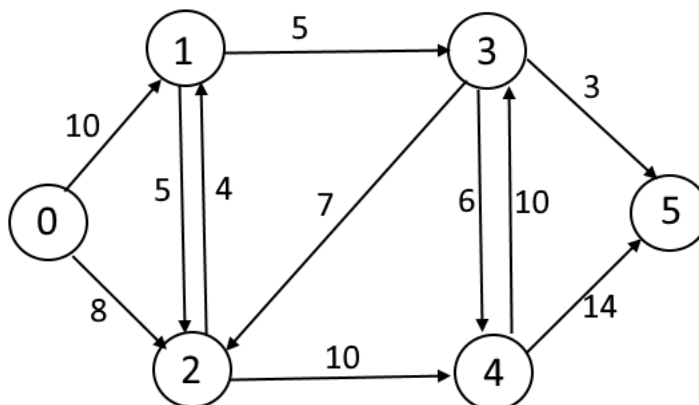
Summer Trimester: 2025

Assignment: 01

Total marks: 05

- 1 (a) An investor plans to save for retirement by making annual deposits in an account. The account earns an interest rate of 9%, compounded monthly. They will start with an initial deposit of \$3,000. And increase each subsequent deposit by \$500 at the end of every year for 5 years. What will be the **future value** of this annuity at the end of 5 years? What will be the **Present value** of this annuity at the end of 5 years?
- (b) A public project is being assessed for feasibility based on projected benefits and costs over a 5-year period. At year 0, the project requires an initial investment of \$30 bn, with no immediate benefits. In year 1, benefits are projected at \$20 bn, while costs are \$10 bn. Year 2 forecasts \$40 bn in benefits with \$10 bn in costs, followed by year 3 with benefits of \$50 bn and costs of \$15 bn. By year 4, benefits are expected to be \$30 bn, with associated costs of \$20 bn. In the final year, both benefits and costs are estimated at \$20 bn. Assuming an interest rate of 8% compounded monthly, compute the Benefit-Cost Ratio (B/C) to evaluate whether the project should be accepted or rejected. (bn= Billion)

2 (a)



Find the maximum flow, here (the maximum allowable limit in written on the line)

- (b) Two projects are given
Project "A"

Year	0	1	2	3	4	5
Cash Flow	-30200	5520	8500	12300	16300	20000

Project "B"

Year	0	1	2	3	4	5
Cash Flow	-25,200	8,900	10,000	6,200	6,600	17,500

Now select the project using the **Discounted payback period method** and **consider the rate = 23% compounded weekly**. Which project should you select?

- 3 (a) Explain what you mean by IRR and how a business can be benefitted by using IRR.

(b)

Year	0	1	2	3	4	5
Project A	-100,000	25,000	35,000	55,000	20,000	15,000
Project B	-200,000	65,000	58,000	55,000	37,000	40,000

Calculate the **IRR**, starting from 10%, using the trial-and-error method. If the required rate of return is **12%**, should the company accept the projects? (You must show the necessary calculations).

- 4 (a) Why is EOQ considered a "trade-off" model? Which two costs does it balance?
- (b) Surge Electric uses 4,000 toggle switches a year. Switches are priced as follows: It costs approximately \$30 to prepare an order and receive it, and carrying costs are 40 percent of the purchase price per unit annually. Determine the optimal order quantity and the total annual cost.

Ordered quantity	Unit price
1 - 499	\$0.90
500 - 999	\$0.85
1000 or more	\$0.80

5 (a)

Task	Predecessors	Duration
A	--	3
B	--	4
C	A,B	5
D	B,C	5
E	A, B,C	7
F	C,E	4
G	E,A,D	5
H	F,G	3

Find the Critical Path.

- (b) Explain each component of supply chain management in details with relevant examples.